

Customer Data Analytics



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Share of Wallet

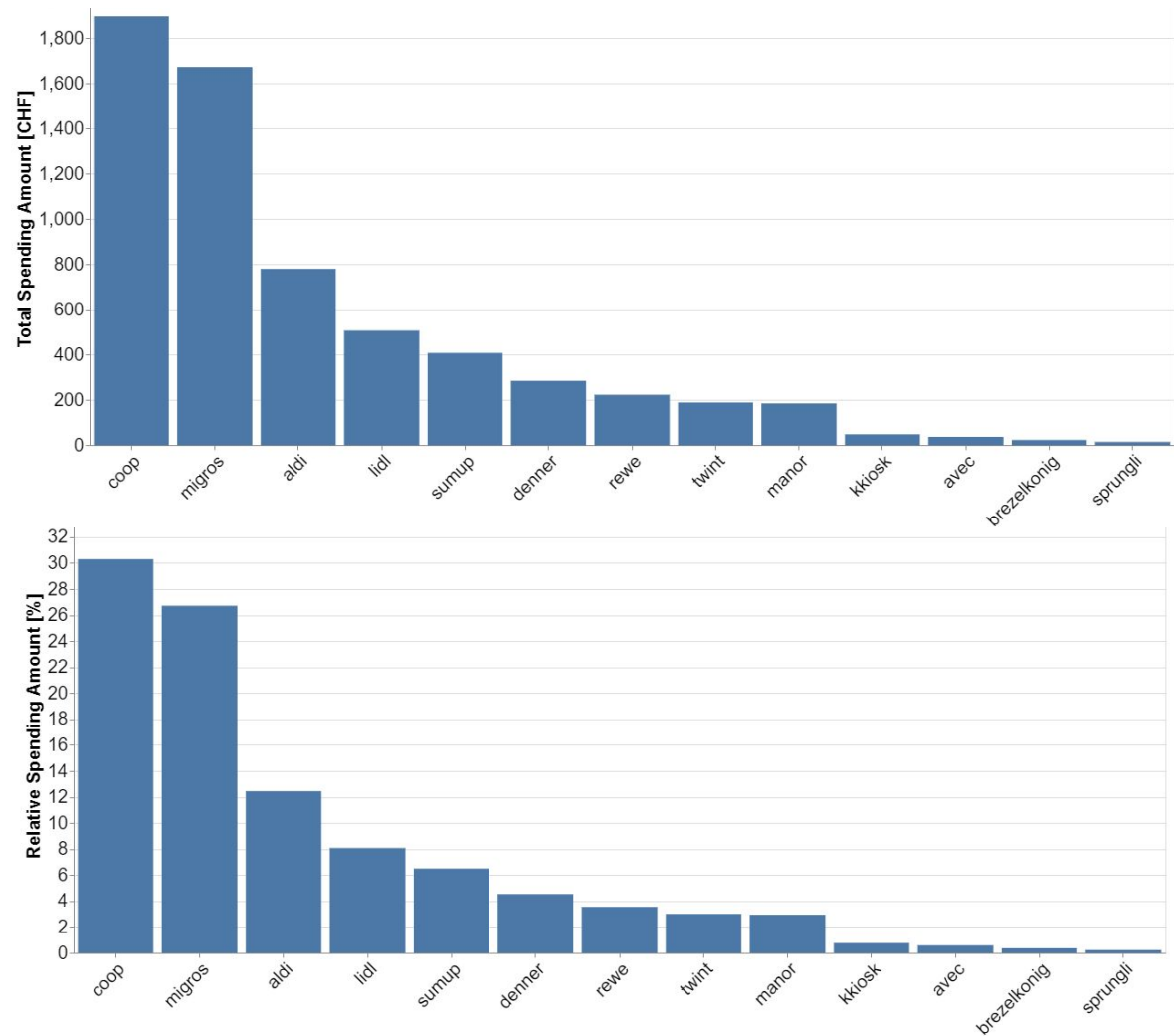
Share of wallet is the percentage ("share") of a customer's expenses ("of wallet") for a product that goes to the firm selling the product. Different firms fight over the share they have of a customer's wallet, all trying to get as much as possible.

https://en.wikipedia.org/wiki/Share_of_wallet

Share of Wallet

Total and relative spending on groceries of a single customer

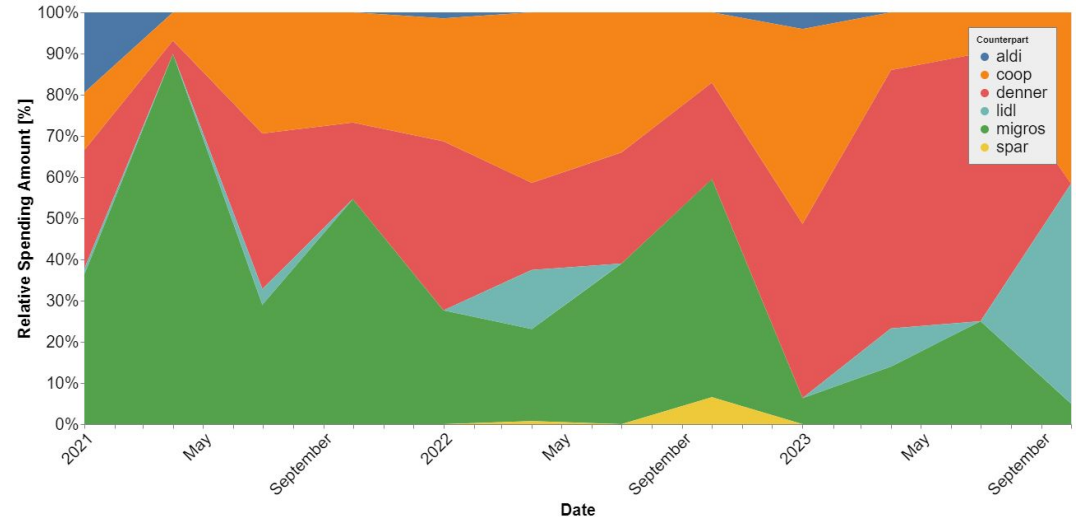
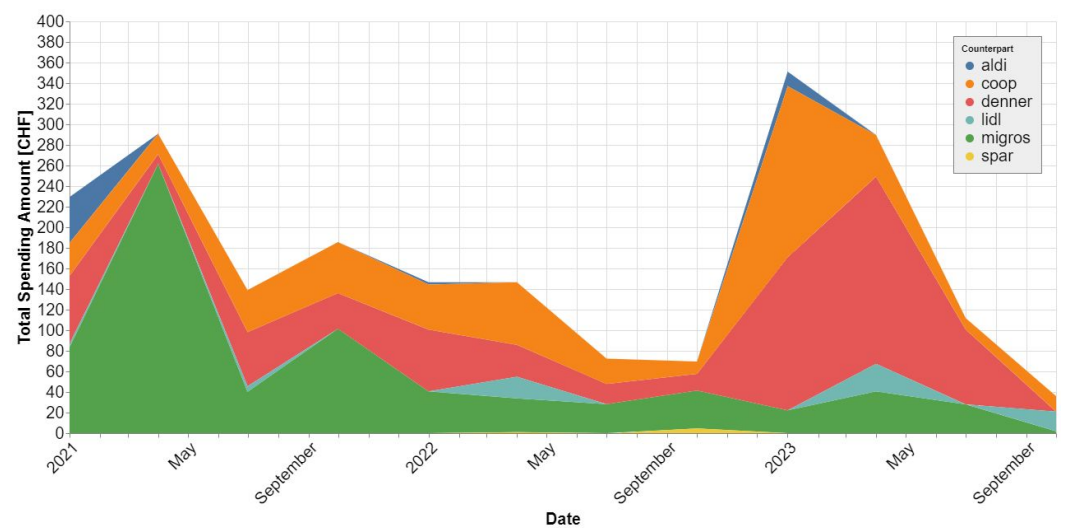
Coop has a SOW of 30.3%, closely followed by Migros with 26.7%



SOW over Time

SOW over time of a single customer. The total spending peaked in March 2021 and again at the beginning of 2023.

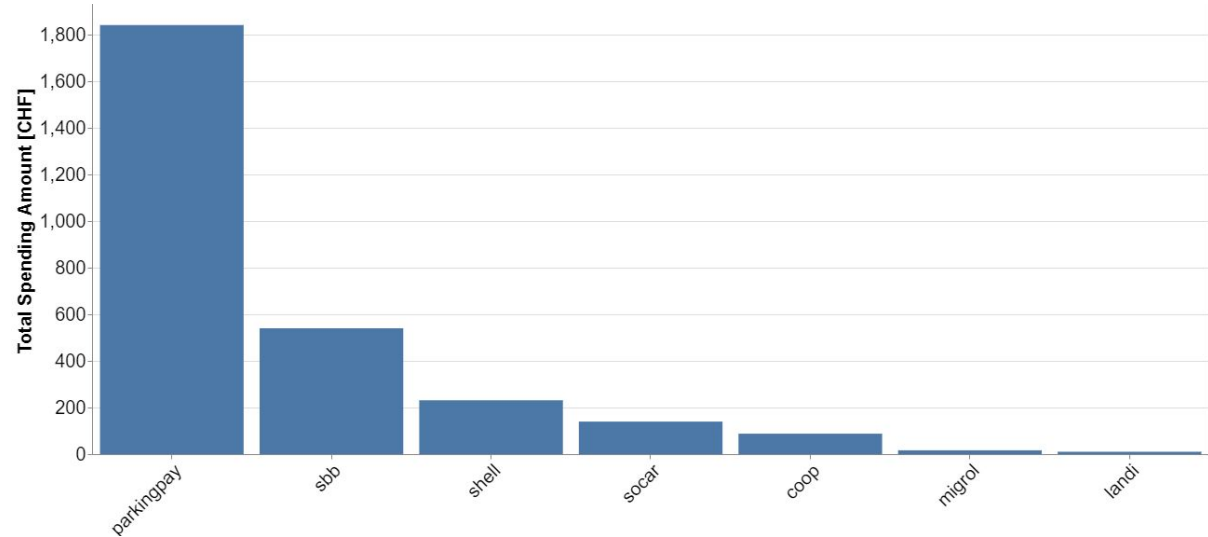
SOW seems to have shifted over time from Migros to Denner and Coop.



Transport Spending

Total spending of a customer in the category “transport”.

This customer seems to be traveling mainly by car and spending money on parking and gas, but still travels by public transport from time to time.



Customer Churn

*The definition of **customer churn** is when a customer decides not to be a customer any longer. In short, they stop buying your products or services.*

*A certain amount of **customer churn** is a natural part of the business cycle as you gain new customers and lose a few old ones for various reasons. Even the best companies lose customers from time to time, often for reasons outside of their control.*

*But too much **churn** is damaging to your business. Acquiring new customers is expensive, and losing lots of previously steady customers indicates potential issues with your products or customer experience.*

<https://uk.surveymonkey.com/mp/how-to-reduce-customer-churn/>

Customer Churn

For a given time frame (month, quarter, year, etc.) we can look at different metrics:

- **Customer Churn** = number of customers lost
- **Churn Rate (%)** = (number of customers lost) / (number of customers at beginning of time frame)

Example:

We have 1000 Customers at the beginning of the year. In the first quarter we lose 50 customers due to various reasons.

Churn Rate = $50 / 1000 = 0.05 = 5\%$

Customer Retention

To identify issues and find out why customers stop using the services of a company we can try to leverage **customer data**:

- Detect which customers are about to churn and try to find out how they behave prior to churning
- Establish the true value of potential loss of these customers. E.g. try prioritizing to keep higher value customer
- Create a retention plan to reduce or even avoid customer churn

The Data

Aggregated SoW Data

1 Dataset with only **Category** aggregation

1 Dataset with **Category and Top Counterpart** aggregation

- Top counterparts selected per category with at least 1000 transactions
- All other counterparts grouped under “other”
- Transactions grouped by month

Aggregated Customer Data

Dataset consisting of 5500 Customers with aggregated statistics of their credit card transactions, including:

- Number of transactions
- Total amount spent [CHF]
- Days active (last transaction - first transaction)
- Ecom / PoS Percentage of spending
- Number of distinct counterparts
- Spending per category
- Spending in different currencies
- Spending in different countries

Churn Labels

For 70% of the customers there is a label if they have churned or not (true/false).

Can you predict the remaining 30%?

Hand in your prediction and we will evaluate your accuracy.

The Challenge

Provide Data Driven Insights

- Define your **key stakeholder** / customer (Yapeal, retail or service company, other)
- Choose one or several key **research topics**
- Prepare the **data** to enable your **analysis**
- Use **advanced analytics** tools such as ML / DL, GenAI etc.
- Build dashboards, interactive visualizations etc.



Share of Wallet

Full Picture

In the examples so far we only looked at individual customers.

For companies it would be interesting to see their SOW for all users or a subset / cluster of users.

History and Future

How did the SOW of a company change over time.

How will it look like in the future and what can a company do to influence it.

Different Categories

How well are companies doing in different categories.

What are the differences between categories.

Customer Insights

Churn

Identify which customers or customer groups are likely to churn.

Are there any signals?

How could we prevent churn?

Clustering

Create clusters of customers, look for valuable insights.

How can we use clusters to improve the customer experience?

Hand in your churn prediction by 14.10 and we will evaluate your results.

What to do

Predict

Estimate how high / low the SOW.

Predict customer churn.

Cluster

Create clusters of customers or companies.

Describe the clusters and what you could do with them.

Visualize

Build strong and interactive visualizations.

Tell a data story and provide data driven insights.

Help the Business

Build data driven recommendations for a company and provide support in business decisions.

These are only examples!

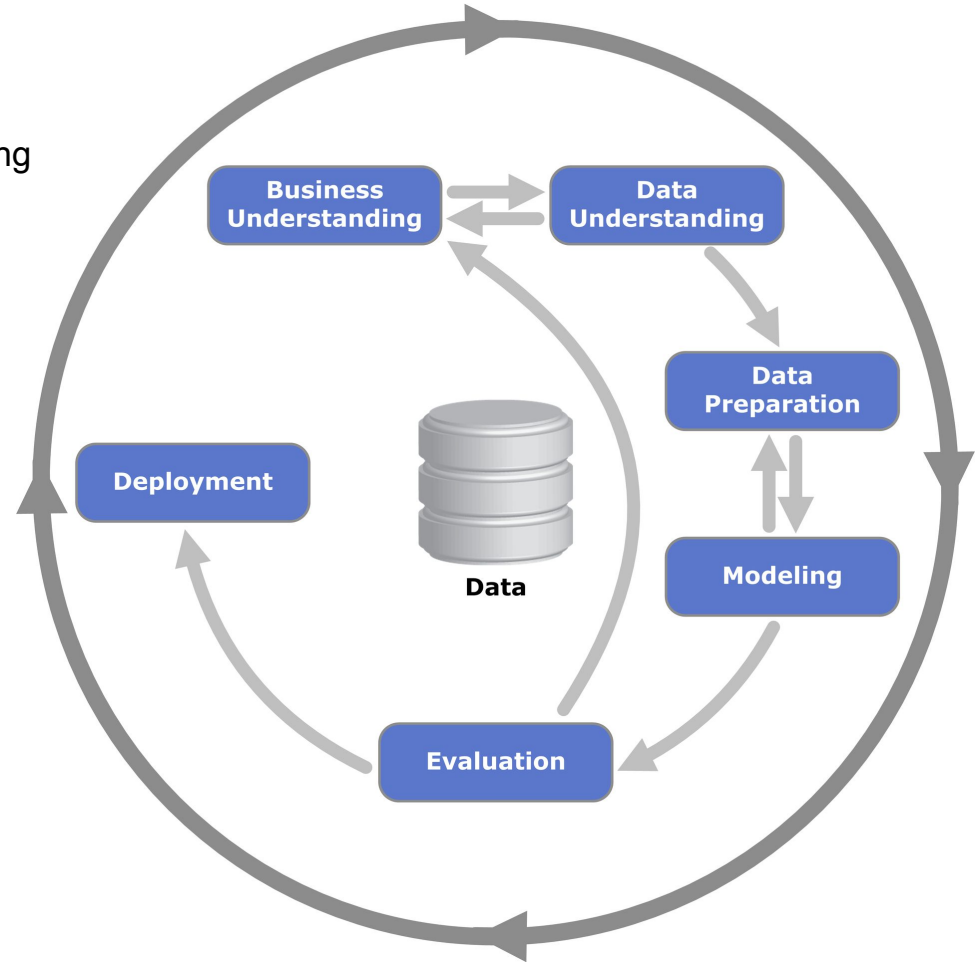
Be Creative!

Process

CRISP-DM

Cross Industry Standard Process for Data Mining

<https://s2.smu.edu/~mhd/8331f03/crisp.pdf>



IBM has released a new framework for Data Mining projects in 2015 called Analytics Solutions Unified Method for Data Mining & Predictive Analytics (ASUM-DM) which is a refined and extended CRISP-DM.

Find Additional Data Sources

OpenData

<https://opendata.swiss/de/>

MeteoSchweiz

<https://www.meteoschweiz.admin.ch/service-und-publikationen/service/open-data.html>

opentransportdata.swiss

<https://opentransportdata.swiss/de/>

Your own

Add any additional data sources
you can think of

Reading Material & Tools

Python for Data Analysis

<https://wesmckinney.com/book/>

Python Data Science Handbook

<https://jakevdp.github.io/PythonDataScienceHandbook/>

Colab online Python Environments

Collaborative coding in a Jupyter style notebook

<https://colab.research.google.com/>

Thank You!

